

Sub Code: 23CE/EE/ME/EC/CS1T05

MIC23

B. Tech I Semester Regular Examinations

**ENGINEERING GRAPHICS**

(Common to CE/EEE/ME/ECE/CSE)

**(Model Paper – II)**

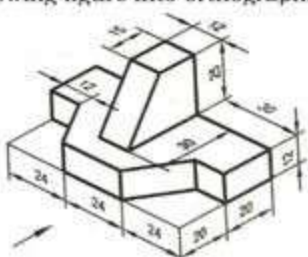
Time: 3 hours

Max Marks: 70M

Note: 1. Answer any five Questions with either or choice. One Question from Each.  
2. All Questions carry Equal Marks. 5 X 14 = 70M

|                   |    |                                                                                                                                                                                                                                                                                                      | CO  | L  | Marks |
|-------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|-------|
| 1.                | a) | Construct a square and hexagon of 40mm side using general construction method.                                                                                                                                                                                                                       | CO1 | L1 | 7 M   |
|                   | b) | Construct a plain scale to show meters when 1 centimeter represents 4 meters and long enough to measure up to 50 meters. Find the R.F. and mark on your scale a distance of 36 meters.                                                                                                               | CO1 | L1 | 7 M   |
| <b>(OR)</b>       |    |                                                                                                                                                                                                                                                                                                      |     |    |       |
| 2.                |    | Draw a cycloid of a circle of diameter 50mm for one revolution. Also draw a tangent & normal to the curve at a point 35mm above the base line                                                                                                                                                        | CO1 | L1 | 14 M  |
| <b>UNIT – II</b>  |    |                                                                                                                                                                                                                                                                                                      |     |    |       |
| 3.                | a) | Draw the Projections of the following points on a common reference line.<br>i) A, 35 mm above HP and 25 mm in front of VP<br>ii) B, 40 mm below HP and 15 mm behind VP<br>iii) C, 50 mm above HP and 25 mm behind VP<br>iv) D, 45 mm below HP and 25 mm behind VP<br>v) E, 30 mm behind VP and on HP | CO2 | L2 | 7 M   |
|                   | b) | A line of 90 long is parallel to and 20 in front of VP. Its one end is in HP. While the other end is 40 above HP. Draw the projections of the line and find its inclination with HP.                                                                                                                 | CO2 | L2 | 7 M   |
| <b>(OR)</b>       |    |                                                                                                                                                                                                                                                                                                      |     |    |       |
| 4.                | a) | A regular pentagon of 30 side is resting on one of its edges on H.P., which is inclined at $45^\circ$ to V.P. its surface is inclined at $30^\circ$ to H.P. Draw its projections.                                                                                                                    | CO2 | L3 | 7 M   |
|                   | b) | A thin $30^\circ - 60^\circ$ set square has its longest edge (diagonal) on H.P and inclined at $30^\circ$ to V.P. its surface makes an angle $45^\circ$ with H.P. Draw the projections, choosing suitable size of the set square                                                                     | CO2 | L3 | 7 M   |
| <b>UNIT – III</b> |    |                                                                                                                                                                                                                                                                                                      |     |    |       |
| 5.                | a) | Draw the projections of a cone 70 mm height and 50 mm diameter, resting on one of its generators on H.P and its axis lies in a vertical plane, inclined at $30^\circ$ to V.P.                                                                                                                        | CO3 | L4 | 7 M   |
|                   | b) | Draw the projections a square prism with base on H.P and a side of the base inclined at $45^\circ$ to VP, taking the side of the base equal to 40 mm and the length of axis as 70 mm                                                                                                                 | CO3 | L4 | 7 M   |
| <b>(OR)</b>       |    |                                                                                                                                                                                                                                                                                                      |     |    |       |
| 6.                | a) | Draw the projections of a cylinder, base 30mm diameter and axis 40mm long, rests with a point of its base circle on H.P. such that the axis is making an angle of $30^\circ$ with H.P.                                                                                                               | CO3 | L4 | 7 M   |
|                   | b) | A pentagonal pyramid side of base 20 mm and axis 45 mm long rests with one of its corners on HP such that the base is inclined at an angle of $60^\circ$ to HP and one side of the base is perpendicular to VP. Draw its projections.                                                                | CO3 | L4 | 7 M   |
| <b>UNIT – IV</b>  |    |                                                                                                                                                                                                                                                                                                      |     |    |       |

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|                                                                                     |    |                                                                                                                                                                                                                                                                                                                                                  |     |    |      |
|-------------------------------------------------------------------------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|------|
| 7.                                                                                  | a) | A cube of 45 mm side rests with a face on HP such that one of its vertical faces is inclined at $30^\circ$ to VP. A section plane, parallel to VP cuts the cube at a distance of 15 mm from the vertical edge nearer to the observer. Draw its top and sectional front views.                                                                    | CO4 | L4 | 7 M  |
|                                                                                     | b) | A pentagonal pyramid side of base 35 mm and axis 60 mm long, rests with its base on HP such that one of the edges of the base is perpendicular to VP. A section plane perpendicular to HP and parallel to VP cuts the pyramid at a distance of 20 mm from the corner of the base nearer to the observer. Draw its top and sectional front views. | CO4 | L4 | 7 M  |
| <b>(OR)</b>                                                                         |    |                                                                                                                                                                                                                                                                                                                                                  |     |    |      |
| 8.                                                                                  | a) | Draw the development of the lateral surface of the cone of base diameter 48 mm and altitude 55 mm.                                                                                                                                                                                                                                               | CO4 | L4 | 7 M  |
|                                                                                     | b) | Draw the development of the complete surface of a G.I. cylindrical drum with lid. Diameter is 30 cm and height is 1.6 times the diameter.                                                                                                                                                                                                        | CO4 | L4 | 7 M  |
| <b>UNIT – V</b>                                                                     |    |                                                                                                                                                                                                                                                                                                                                                  |     |    |      |
| 9.                                                                                  | a) | Convert the following figure into orthographic Views                                                                                                                                                                                                                                                                                             | CO5 | L4 | 14 M |
|  |    |                                                                                                                                                                                                                                                                                                                                                  |     |    |      |
| 10.                                                                                 | a) | Convert the following orthographic views to Isometric View                                                                                                                                                                                                                                                                                       | CO5 | L6 | 14 M |

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Sub Code:23CE/EE/ME/ EC/CS1T05

MIC23

B. Tech I Semester Regular Examinations

**ENGINEERING GRAPHICS**

(CE, EEE, MECH, ECE, CSE)

(Model Paper - I)

Time: 3 hours

Max Marks: 70M

Note: 1. Answer any five Questions with either or choice. One Question from Each  
2. All Questions carry Equal Marks. 5 X 14 = 70M

| UNIT - I   |    |                                                                                                                                                                                                                                                                                                                                               | CO  | BTL | Marks |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|
| 1.         | a) | Construct a regular hexagon of side 32mm when one side is vertical.                                                                                                                                                                                                                                                                           | CO1 | L1  | 7 M   |
|            | b) | Draw an involute of a pentagon of side 25mm.                                                                                                                                                                                                                                                                                                  | CO1 | L2  | 7 M   |
| (OR)       |    |                                                                                                                                                                                                                                                                                                                                               |     |     |       |
| 2.         |    | The major axis of an ellipse is 100mm and the minor axis is 55mm. Find the foci and construct the ellipse by Intersecting Arcs Method.                                                                                                                                                                                                        | CO1 | L2  | 14 M  |
| UNIT - II  |    |                                                                                                                                                                                                                                                                                                                                               |     |     |       |
| 3.         | a) | A line of 100 long is parallel to and 30 above HP. Its two ends are 25 and 50 in front of VP respectively. Find its inclination with VP.                                                                                                                                                                                                      | CO2 | L3  | 7 M   |
|            | b) | A line AB 65mm long has its end A 20mm above H.P and 25mm in front of V.P. the end B is 40mm above H.P. and 65mm in front of V.P. Draw its projections and find its inclinations with the HP and VP. Mention the values of apparent angles.                                                                                                   | CO2 | L3  | 7 M   |
| (OR)       |    |                                                                                                                                                                                                                                                                                                                                               |     |     |       |
| 4.         | a) | A thin rectangular plate 60x40 size has its shorter edge on H.P and inclined at 30° to V.P. draw the projections of the plate, when its top view is a square of 40 side.                                                                                                                                                                      | CO2 | L4  | 7 M   |
|            | b) | Mark the Projections of the following points on a common reference line, keeping the projectors 35 mm apart.<br>i) A, 25 mm above HP and 35 mm in front of VP.<br>ii) B, 25 mm above HP and 40 mm behind VP<br>iii) C, 30 mm below HP and 45 mm behind VP<br>iv) D, 30 mm below HP and 40 mm in front of VP<br>v) E, 25 mm above HP and on VP | CO2 | L3  | 7 M   |
| UNIT - III |    |                                                                                                                                                                                                                                                                                                                                               |     |     |       |
| 5.         | a) | Draw the projections a pentagonal prism with a rectangular face on H.P, axis perpendicular V.P and one base on V.P, taking the side of the base equal to 35 mm and the length of axis as 60 mm                                                                                                                                                | CO3 | L4  | 7 M   |
|            | b) | Draw the projections of a cone, base 30 mm diameter and axis 50 mm long, resting on HP on a point of its base circle with the axis making an angle of 45° with HP and parallel to VP.                                                                                                                                                         | CO3 | L4  | 7 M   |
| (OR)       |    |                                                                                                                                                                                                                                                                                                                                               |     |     |       |
| 6.         | a) | A hexagonal pyramid side of base 25mm and axis 50mm long rests with one of the edges of its base on H.P. and its axis is inclined at 30° to H.P and parallel to V.P. Draw its projections.                                                                                                                                                    | CO3 | L4  | 7 M   |
|            | b) | Draw the projections of the cylinder, with its axis                                                                                                                                                                                                                                                                                           | CO3 | L4  | 7 M   |

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|           |    | perpendicular to V.P and 40 mm above H.P and one end at 20 mm from V.P., taking the diameter of the base equal to 50 mm and the length of the axis is 60 mm.                                                                                                                                                                                                                                  |     |    |      |
|-----------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|------|
| UNIT - IV |    |                                                                                                                                                                                                                                                                                                                                                                                               |     |    |      |
| 7.        | a) | A hexagonal prism, side of base 30mm and axis 60 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. A section plane perpendicular to HP and parallel to VP cuts the prism at a distance of 10 mm from its axis. Draw its top and sectional front views.                                                                                             | CO4 | L4 | 7 M  |
|           | b) | A square prism, side of base 30 mm and axis 60 mm long, rests with its base on HP such that one of its rectangular faces is inclined at 30° to VP. A section plane perpendicular to HP and inclined at 60° to VP passes through the prism such that a rectangular face which is making 60° with VP is cut into two halves. Draw the top view, sectional front view and true shape of section. | CO4 | L4 | 7 M  |
| (OR)      |    |                                                                                                                                                                                                                                                                                                                                                                                               |     |    |      |
| 8.        | a) | Draw the development of the lateral surface of a right square prism of edge of base 30mm and axis 50 mm long                                                                                                                                                                                                                                                                                  | CO4 | L3 | 7 M  |
|           | b) | Draw the development of the lateral surfaces of a square pyramid, side of base 25 mm and height 50 mm, resting with its base on HP and an edge of the base parallel to VP.                                                                                                                                                                                                                    | CO4 | L3 | 7 M  |
| UNIT - V  |    |                                                                                                                                                                                                                                                                                                                                                                                               |     |    |      |
| 9.        |    | Convert the following figure into orthographic Views                                                                                                                                                                                                                                                                                                                                          |     |    |      |
|           |    |                                                                                                                                                                                                                                                                                                            | CO5 | L6 | 14 M |

Date: 14 FEB 2024

Regd. No.

Sub Code: 23CE/EE/ME/EC1T05

MIC23

B. Tech I Semester Regular Examinations, Jan-2024

**ENGINEERING GRAPHICS**

(CE, EEE, ME, ECE)

Time: 3 hours

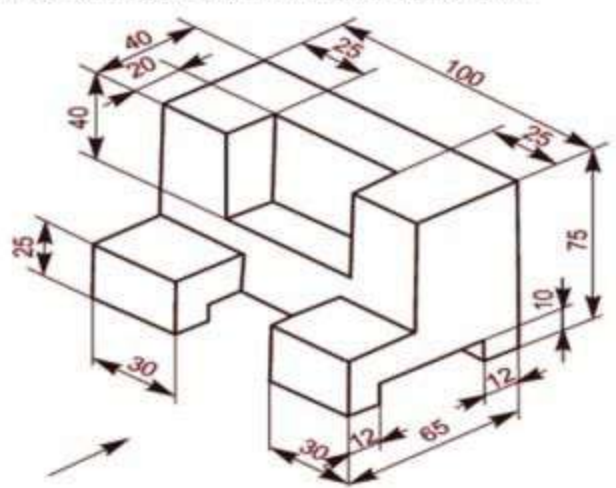
Max Marks: 70M

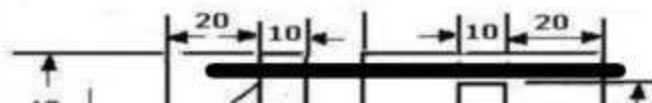
Note: 1. Answer any five Questions with either or choice. One Question from Each.  
2. All Questions carry Equal Marks. 5 X 14 = 70M

|            |    |                                                                                                                                                                                                                                                                                            | Marks | L  | CO  |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|-----|
| 1.         | a) | Construct a regular hexagon of side 32mm when one side is vertical.                                                                                                                                                                                                                        | 7M    | L1 | CO1 |
|            | b) | Draw an involute of a pentagon of side 25mm.                                                                                                                                                                                                                                               | 7M    | L2 | CO1 |
| (OR)       |    |                                                                                                                                                                                                                                                                                            |       |    |     |
| 2.         |    | Construct an ellipse, with distance of the focus from the directrix as 50 mm and eccentricity as $2/3$ . Also, draw normal and tangent to the curve at a point 40 mm from the directrix                                                                                                    | 14M   | L2 | CO1 |
| UNIT - II  |    |                                                                                                                                                                                                                                                                                            |       |    |     |
| 3.         |    | Draw the projections of the following points on the same reference line. Keeping the projectors 40 mm apart.<br>(i) A, 40 mm below HP and 30 mm in front of VP.<br>(ii) B, 45 mm below HP and 30 mm behind VP.<br>(iii) C, 45 mm below HP and in VP.<br>(iv) D, 20 mm behind VP and in HP. | 14M   | L2 | CO2 |
| (OR)       |    |                                                                                                                                                                                                                                                                                            |       |    |     |
| 4.         |    | A regular hexagonal plane with a 40 mm side rests on a corner in the H.P., and its surface is inclined at $45^\circ$ to the H.P. Draw its projections when the diagonal through the corner, which is on H.P., makes $30^\circ$ with VP.                                                    | 14M   | L3 | CO2 |
| UNIT - III |    |                                                                                                                                                                                                                                                                                            |       |    |     |
| 5.         |    | A square prism, base 25 mm side and height 50 mm has its axis inclined at $45^\circ$ to H.P and has an edge of its base, on H.P and inclined at $30^\circ$ to the V.P. Draw its projections.                                                                                               | 14M   | L2 | CO3 |
| (OR)       |    |                                                                                                                                                                                                                                                                                            |       |    |     |
| 6.         |    | Draw the projections a pentagonal prism with a rectangular face on H.P, axis perpendicular V.P and one base on V.P, taking the side of the base equal to 35 mm and the length of axis as 60 mm.                                                                                            | 14M   | L3 | CO3 |
| UNIT - IV  |    |                                                                                                                                                                                                                                                                                            |       |    |     |
| 7.         |    | A hexagonal pyramid of side 30 mm and axis height 70 mm is resting on H.P being one side of the base parallel to the V.P. Draw the development of the lateral surface of the solid.                                                                                                        | 14M   | L3 | CO4 |
| (OR)       |    |                                                                                                                                                                                                                                                                                            |       |    |     |
| 8.         |    | A pentagonal pyramid of side 30 mm and axis height 70 mm is resting on H.P being one side of the base parallel to the V.P. Draw the development of the lateral surface of the solid.                                                                                                       | 14M   | L2 | CO4 |

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## UNIT - V

|      |                                                                                                                       |     |    |     |
|------|-----------------------------------------------------------------------------------------------------------------------|-----|----|-----|
| 9.   | Draw the front view, top view and the right side view of the drawing given in figure below. All the dimensions in mm. |     |    |     |
|      |                                    | 14M | L4 | CO5 |
| (OR) |                                                                                                                       |     |    |     |

|     |                                                                                     |  |  |  |
|-----|-------------------------------------------------------------------------------------|--|--|--|
| 10. | Convert the following orthographic views to Isometric View                          |  |  |  |
|     |  |  |  |  |



Date: 14 FEB 2024

Regd. No.

Sub Code: 23CS1T05

MIC23

B. Tech I Semester Regular Examinations, Jan-2024

**ENGINEERING GRAPHICS**

(CSE)

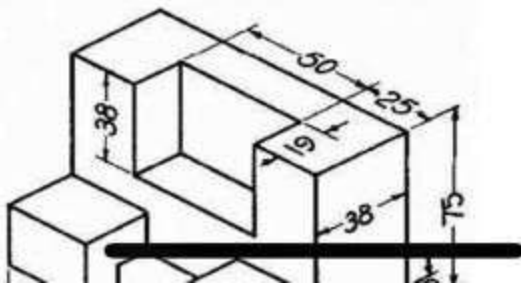
Time: 3 hours

Max Marks: 70M

Note: 1. Answer any five Questions with either or choice. One Question from Each.  
2. All Questions carry Equal Marks. 5 X 14 = 70M

|            |    |                                                                                                                                                                                                                                                                                   | Marks | BL | CO  |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|-----|
| 1.         | a) | Inscribe a regular polygon of sides, say five, in a circle of 64 mm diameter when one side is horizontal.                                                                                                                                                                         | 7M    | L1 | CO1 |
|            | b) | Construct a diagonal scale to read 2 km when its R.F = 1 : 20000. Mark on it a distance of 1.15 km.                                                                                                                                                                               | 7M    | L1 | CO1 |
| (OR)       |    |                                                                                                                                                                                                                                                                                   |       |    |     |
| 2.         |    | Draw an epicycloid having a generating circle of diameter 75 mm and a directing curve of radius 200 mm. Also draw a normal and a tangent at a point P on the curve.                                                                                                               | 14M   | L1 | CO1 |
| UNIT - II  |    |                                                                                                                                                                                                                                                                                   |       |    |     |
| 3.         | a) | A point A is on H.P and 40 mm in front of V.P. Another point B is on V.P and below H.P. The line joining their front views makes an angle of $45^\circ$ with xy, while the line joining their top views makes an angle of $30^\circ$ . Find the distance of the point B from H.P. | 7M    | L2 | CO2 |
|            | b) | A line MN measures 70 mm long is 25 mm above HP and 35 mm in front of VP. The line is inclined $35^\circ$ to HP and $45^\circ$ to VP. Draw its projections.                                                                                                                       | 7M    | L2 | CO2 |
| (OR)       |    |                                                                                                                                                                                                                                                                                   |       |    |     |
| 4.         | a) | A semi - circular plate of 80 mm diameter, has its straight edge on V.P and inclined at $30^\circ$ to H.P. Draw its projections.                                                                                                                                                  | 7M    | L3 | CO2 |
|            | b) | Draw the projections of a regular hexagon of 25 mm side having one of its sides in H.P and inclined at $60^\circ$ to V.P. and its surface making an angle of $45^\circ$ with H.P.                                                                                                 | 7M    | L3 | CO2 |
| UNIT - III |    |                                                                                                                                                                                                                                                                                   |       |    |     |
| 5.         | a) | A cone 40 mm base and axis 50 mm long touches to V.P on a point of its base circle. The axis inclined at $30^\circ$ to V.P and front view of the axis is inclined at $45^\circ$ to H.P. Draw its projections.                                                                     | 7M    | L4 | CO3 |
|            | b) | Draw the projections of a square prism, side of base 30 mm and axis 60 mm long resting with one of the edge of its base on H.P. Its axis is inclined at $30^\circ$ to H.P and the top view of the axis at $45^\circ$ to xy line.                                                  | 7M    | L4 | CO3 |
| (OR)       |    |                                                                                                                                                                                                                                                                                   |       |    |     |
| 6.         | a) | Draw the projections of pentagonal prism of base 25 mm and axis 50 mm long is resting on one of its rectangular faces on H.P. The axis of the solid is inclined at $45^\circ$ to V.P.                                                                                             | 7M    | L4 | CO3 |
|            | b) | A hexagonal pyramid, side of base 25 mm and axis 50 mm long, rests with one of the edges of its base on H.P and its axis is inclined at $30^\circ$ to H.P and parallel to V.P. Draw its projections.                                                                              | 7M    | L4 | CO3 |

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| UNIT - IV |  |                                                                                                                                                                                                                                                                                                                                           |     |    |     |
|-----------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|-----|
| 7.        |  | A cone with diameter of base 50 mm and axis 60 mm long, is resting on its base on H.P. It is cut by section plane inclined at $45^\circ$ to H.P and passing through the axis at a point 35 mm above H.P. Draw the projections of the cut solid.                                                                                           | 14M | L4 | CO4 |
| (OR)      |  |                                                                                                                                                                                                                                                                                                                                           |     |    |     |
| 8.        |  | A cylinder of diameter of base 40 mm and axis 55 mm long, is resting on its base on H.P. It is cut by a section plane, perpendicular to V.P and inclined at $45^\circ$ to H.P. The section plane is passing through the top end of an extreme generator of the cylinder. Draw the development of the lateral surface of the cut cylinder. | 14M | L4 | CO4 |
| UNIT - V  |  |                                                                                                                                                                                                                                                                                                                                           |     |    |     |
| 9.        |  | Convert the following figure into orthographic Views                                                                                                                                                                                                                                                                                      |     |    |     |
|           |  |                                                                                                                                                                                                                                                        | 14M | L4 | CO5 |

Date: 03 JAN 2025

Regd. No.

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Sub Code: 23CS1T05

MIC23

B. Tech I Semester Regular/Supplementary Examinations, Dec-2024

**ENGINEERING GRAPHICS**

(CSE)

Time: 3 hours

Max Marks: 70M

Note: 1. Answer any five Questions with either or choice. One Question from Each.  
2. All Questions carry Equal Marks. 5 X 14 = 70M

| UNIT - I   |    |                                                                                                                                                                                                                                                        | CO  | BTL | Marks |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|
| 1.         | a) | Write freehand the following, using single stroke vertical letters. <b>LETTERING SHOULD BE DONE FREEHAND WITH SPEED</b> (h = 5 mm)                                                                                                                     | CO1 | L1  | 7M    |
|            | b) | The distance between the directrix and the focus of a parabola is 50 mm. Draw the curve.                                                                                                                                                               | CO1 | L2  | 7M    |
| (OR)       |    |                                                                                                                                                                                                                                                        |     |     |       |
| 2.         | a) | Inscribe a square in a circle of diameter 50 mm.                                                                                                                                                                                                       | CO1 | L1  | 7M    |
|            | b) | Construct a plain scale of convenient length to measure a distance of 1 cm and mark on it a distance of 0.94 cm.                                                                                                                                       | CO1 | L2  | 7M    |
| UNIT - II  |    |                                                                                                                                                                                                                                                        |     |     |       |
| 3.         | a) | Draw the Projections of the following points on a common reference line.<br>i) Point A is 35 mm below HP and 25 mm behind VP<br>ii) Point B is 40 mm behind VP and 30 mm above HP<br>iii) Point C is 50 mm above HP and 40 mm in front of VP           | CO2 | L1  | 7M    |
|            | b) | line AB 40 mm long is parallel to VP and inclined at an angle of $30^\circ$ to HP. The end A is 15 mm above HP and 20 mm in front of VP. Draw the projections of the line.                                                                             | CO2 | L2  | 7M    |
| (OR)       |    |                                                                                                                                                                                                                                                        |     |     |       |
| 4.         |    | A regular hexagon plane of 25 side is resting on one of its edges on H.P, which is inclined at $45^\circ$ to V.P. its surface is inclined at $30^\circ$ to H.P. Draw its projections.                                                                  | CO2 | L2  | 14M   |
| UNIT - III |    |                                                                                                                                                                                                                                                        |     |     |       |
| 5.         | a) | A square pyramid, base 40 mm side and axis 65 mm long, has its base in the V.P. One edge of the base is inclined at $30^\circ$ to the H.P and a corner contained by that edge is on the H.P. Draw its projections.                                     | CO3 | L2  | 7M    |
|            | b) | Draw the projections of a triangular prism, base 40 mm side and axis 50 mm long, resting on one of its bases on the H.P. with a vertical face perpendicular to the V.P                                                                                 | CO3 | L2  | 7M    |
| (OR)       |    |                                                                                                                                                                                                                                                        |     |     |       |
| 6.         |    | A cube of 50 mm side rests with a face on HP such that one of its vertical faces is inclined at $40^\circ$ to VP. Draw its top and front views.                                                                                                        | CO3 | L2  | 14M   |
| UNIT - IV  |    |                                                                                                                                                                                                                                                        |     |     |       |
| 7.         |    | A cylinder of base diameter 50 mm and height 65mm rests on its base on HP. It is cut by a plane perpendicular to VP and inclined at $30^\circ$ to HP and meets the axis at a distance 30 mm from the base. Draw the front view and sectional top view. | CO4 | L2  | 14M   |

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| (OR)     |    |                                                                                      |     |    |     |
|----------|----|--------------------------------------------------------------------------------------|-----|----|-----|
| 8.       | a) | Draw the development of a square prism with side of base 30 mm and height 50 mm.     | CO4 | L2 | 7M  |
|          | b) | Draw the development of cylinder of diameter 40 mm and height 60 mm.                 | CO4 | L2 | 7M  |
| UNIT - V |    |                                                                                      |     |    |     |
| 9.       |    | Draw the front and top view of the object shown in figure. All dimensions are in mm. | CO5 | L2 | 14M |
|          |    |                                                                                      |     |    |     |